Chapter 16 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

C 1. Which one of the following statements regarding K_w is false?
A) K_w changes with temperature.
B) pK_w is 14.00 at 25°C.
C) The value of K_w is always 1.0×10^{-14}.
D) The value of K_w shows that water is a weak acid.
E) K_w is known as the ion product of water.

$D_2O \rightleftharpoons D^+ + DO^-$.

B. 2. Deuterium oxide (D_2O, where D is deuterium, the hydrogen-2 isotope) has an ion-product constant, K_w, of 8.9×10^{-16} at 20 °C.The $[D^+]$ for pure (neutral) D_2O at this temperature is _____ *M*.
A) 7.52 B) 3.0×10^{-8} C) 1.0×10^{-14} D) 1.0×10^{-7} E) 7.00

A. 3. HA is a weak acid. Which equilibrium corresponds to the equilibrium constant K_b for A^-?
A) A^- (aq) + H_2O (l) ⇌ HA (aq) + OH^- (aq)
B) HA (aq) + H_2O (l) ⇌ H_2A^+ (aq) + OH^- (aq)
C) A^- (aq) + OH^- (aq) ⇌ HOA^{2-} (aq)
D) A^- (aq) + H_3O^+ (aq) ⇌ HA (aq) + H_2O (l)
E) HA (aq) + H_2O (l) ⇌ H_3O^+(aq) + A^- (aq)

D 4. Is it possible for a solution to have a negative pH? If so, would that pH signify a basic or acidic solution?
A) No, the pH range is 1~14.
B) No, the definition of pH does not permit it to have a negative value.
C) Yes, for any solution with a concentration of base greater than 1 *M*.
D) Yes, for any solution with a concentration of acid greater than 1 *M*.
E) Yes, for any solution with a concentration of acid smaller than 1 *M*.

$\frac{10^{-14}}{2.5 \times 10^{-4}}$ 2.5×10^{-3} 4×10^{-11}

B. 5. What is the pOH of an aqueous solution at 25.0 °C that contains 2.5×10^{-4} *M* hydronium ion? $-\log[OH^-]$
A) 4.0×10^{-11} B) 10.40 C) 9.40 D) 3.60 E) 7.00

A. 6. Determine the pH of a 0.20 *M* aqueous solution of KF at 25.0 °C. For hydrofluoric acid, $K_a = 7.0 \times 10^{-4}$ at 25.0 °C.
A) 8.23 B) 4.23 C) 5.77 D) 10.23 E) 9.77

$F^- + H_2O \rightleftharpoons HF + OH^-$. $\frac{x^2}{0.2-x} = K_b = \frac{[HF][OH^-]}{[F^-]} = \frac{[HF][OH^-][H^+]}{[H^+][F^-]}$

0.2-x x x 0.2

diminished.

B. 7. The acid-dissociation constants of sulfurous acid (H_2SO_3) are $K_{a1} = 1.7 \times 10^{-2}$ and $K_{a2} = 6.4 \times 10^{-8}$ at 25.0 °C. Calculate the pH of a 0.163 *M* aqueous solution of sulfurous acid. $= \frac{K_w}{K_a}$
A) 4.53 B) 1.35 C) 1.86 D) 6.21 E) 1.28

$K_{a1}K_{a2} = \frac{[H^+][HSO_3^-]}{[H_2SO_3]} \frac{[H^+][SO_3^{2-}]}{[HSO_3^-]}$

C. 8. Consider the hydrogen halides HF, HCl, HBr and HI. Which of the statements about them is true?
A) They are all strong acids.
B) They are all weak acids.
C) The boiling point increases with molar mass.
D) The bond energy increases with molar mass.
E) none of above

$H_2SO_3 \rightleftharpoons H^+ + HSO_3^-$. $\frac{(x+y)(x-y)}{0.163-x} = 1.7 \times 10^{-2}$.
0.163-x x+y x-y
$HSO_3^- \rightleftharpoons H^+ + SO_3^{2-}$. $\frac{(x+y)\,y}{x-y} = 6.4 \times 10^{-8}$.
x-y x+y y

A. 9. The dihydrogenphosphate ion, $H_2PO_4^-$ is amphiprotic. In which of the following reactions is/are this ion serving as a Brønsted-Lowry base?
I. $H_3O^+(aq) + H_2PO_4^-(aq) \rightleftharpoons H_3PO_4(aq) + H_2O(l)$
II. $H_3O^+(aq) + HPO_4^{2-}(aq) \rightleftharpoons H_2PO_4^-(aq) + H_2O(l)$
III. $H_3PO_4(aq) + HPO_4^{2-}(aq) \rightleftharpoons 2\ H_2PO_4^-(aq)$

$\frac{x^2}{0.163-x} = 1.7 \times 10^{-2} \times 6.4 \times 10^{-8}$.

A) I only B) I and II C) I and III D) II and III E) All are correct.

D. 10. In which of the following solutions is the percent ionization of the acid the greatest?

Weak acid	K_a
HBrO	2.5×10^{-9}
$HC_3H_5O_2$	1.3×10^{-5}

$[H^+] \approx \sqrt{K_a \cdot C_o}$ $C_o \downarrow$

$\% \approx \sqrt{\frac{K_a}{C_o}}$ $\% \uparrow$.

A) 0.10 *M* HBrO(*aq*)
B) 0.10 *M* $HC_3H_5O_2$(*aq*)
C) 0.025 *M* HBrO(*aq*)
D) 0.025 *M* $HC_3H_5O_2$(*aq*)
E) All of the above solutions has the same percent ionization.

B 11. In the gas phase reaction below, NH_3 is acting as a(n) ___ base but not as a(n) ___ base.

$$NH_3 + H^+ \rightleftharpoons NH_4^+$$

A) Arrhenius, Brønsted-Lowry
B) Lewis, Arrhenius
C) Brønsted-Lowry, Lewis
D) Arrhenius, Lewis
E) Lewis, Brønsted-Lowry

E 12. Arrange the following substances in order from weakest to strongest acid:

$HClO_4$, HIO_2, $HBrO_3$, $HClO_3$, HIO_3

A) $HIO_2 < HClO_3 < HBrO_3 < HIO_3 < HClO_4$
B) $HIO_3 < HIO_2 < HClO_4 < HBrO_3 < HClO_3$
C) $HClO_4 < HClO_3 < HBrO_3 < HIO_3 < HIO_2$
D) $HBrO_3 < HIO_3 < HClO_3 < HIO_2 < HClO_4$
E) $HIO_2 < HIO_3 < HBrO_3 < HClO_3 < HClO_4$

E 13. Of the following substances, the aqueous solutions of _____ is/are basic.

NaHS $Cu(NO_3)_2$ $KHCO_3$ NaF

Formula	K_{a1}	K_{a2}
H_2S	9.5×10^{-8}	1×10^{-19}
H_2CO_3	4.3×10^{-7}	5.6×10^{-11}

$HS^- \rightleftharpoons H^+ + S^{2-}$. 1×10^{-19}

$HS^- + H_2O \rightleftharpoons H_2S + OH^-$

$\frac{[H_2S][OH^-][H^+]}{[HS^-][H^+]}$ $\frac{K_w}{K_{a1}}$ $\frac{10^{-14}}{10^{-7}}$

10^{-7}.

$\frac{10^{-14}}{4\times10^{-7}}$ 10^{-7} 10^{-11}

A) NaHS, $Cu(NO_3)_2$
B) $KHCO_3$, NaHS
C) NaF only
D) NaF, $KHCO_3$
E) NaHS, $KHCO_3$ and NaF

E 14. Which one of the following is a Brønsted–Lowry acid?
A) $(CH_3)_3NH^+$ B) CH_3COOH C) HF D) HNO_2 E) all of the above

$\frac{x^2}{0.015-x} = 3\times10^{-8}$.

D 15. The K_a of hypochlorous acid (HClO) is 3.0×10^{-8} at 25 °C. What is the percent ionization(%) of hypochlorous acid in a 0.015 *M* aqueous solution of HClO at 25 °C?
A) 4.5×10^{-8} B) 14 C) 2.1×10^{-5} D) 0.14 E) 0.0014

$\frac{x}{0.015}$

Chapter 16 Homework

Part B Short questions (Write legibly and show all work for all steps in the problem)

16. (12 points)

(a) Write the chemical equation and the K_a expression for the ionization of HCOOH in aqueous solution.

(b) Calculate the pH of a 0.25 *M* solution of HCOOH with $pK_a = 3.74$ at 25 °C.

(c) Calculate the pH of a 0.20 *M* aqueous solution of HCOONa at 25 °C.

(d) The acid-dissociation constants for phosphoric acid (H_3PO_4) are $K_{a1} = 7.5 \times 10^{-3}$, $K_{a2} = 6.2 \times 10^{-8}$, and $K_{a3} = 4.2 \times 10^{-13}$. Is the solution with 1 *M* NaH_2PO_4 acidic, basic, or neutral? Explain.

17. (3 points) The following diagrams represent aqueous solutions of two monoprotic acids, HA(A = X or Y). The water molecules have been omitted for clarity.

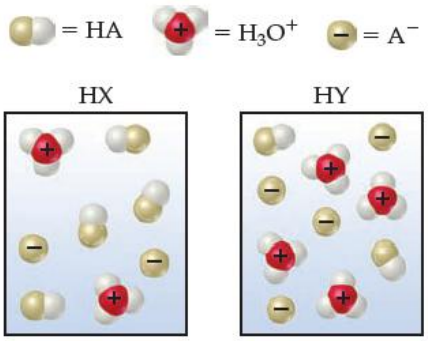
(a) Which is the stronger acid, HX or HY?

(b) Which is the stronger base, X^- or Y^-?

(c) If you mix equal concentrations of HX and NaY, will the equilibrium

$$HX(aq) + Y^-(aq) \rightleftharpoons HY(aq) + X^-(aq)$$

lie mostly to the right ($K_c > 1$) or to the left ($K_c < 1$)?



Chapter 16 Homework Answer Sheet

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Part A multiple choice (15 points)

1-5 CBADB

6-10 ABCAD

11-15 BEEED.

Part B short questions (15 points)

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(a) $HCOOH + H_2O \rightleftharpoons H_3O^+ + HCOO^-$

$$K_a = \frac{[H_3O^+][HCOO^-]}{[HCOOH]}$$

(b) $HCOOH + H_2O \rightleftharpoons H_3O^+ + HCOO^-$

$0.25-x \qquad x \qquad x$

$$\frac{x^2}{0.25-x} = 10^{-3.74}$$

$0.25-x \approx 0.25$

$pH = -\log x = 2.17$

(c)

$HCOO^- + H_2O \rightleftharpoons HCOOH + OH^-$

$0.2-x \qquad x \qquad x$

$$K_b = \frac{[HCOOH][OH^-]}{[HCOO^-]} = \frac{[HCOOH][OH^-][H^+]}{[HCOO^-][H^+]} = \frac{K_w}{K_a} = 10^{-10.26}$$

$x \ll 0.2$

$0.2 \approx 0.2-x$

$$\frac{x^2}{0.2-x} = 10^{-10.26} \qquad pH = -\log \frac{10^{-14}}{x} = 8.52$$

(d). $H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$. $K_{a1} = 7.5\times10^{-3}$.

$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$. $K_{a2} = 6.2\times10^{-8}$.

$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$. $K_{a3} = 4.2\times10^{-13}$.

$H_2PO_4^- + H_2O \rightleftharpoons H_3PO_4 + OH^-$. $K_b = \frac{K_w}{K_{a1}}$.

$K_{a2} \gg K_b \gg K_{a3}$. $\Rightarrow$ acidic.

17. (3 points) The following diagrams represent aqueous solutions of two monoprotic acids, HA(A = X or Y). The water molecules have been omitted for clarity.

(a) Which is the stronger acid, HX or HY?

(b) Which is the stronger base, X^- or Y^-?

(c) If you mix equal concentrations of HX and NaY, will the equilibrium

$$HX(aq) + Y^-(aq) \rightleftharpoons HY(aq) + X^-(aq)$$

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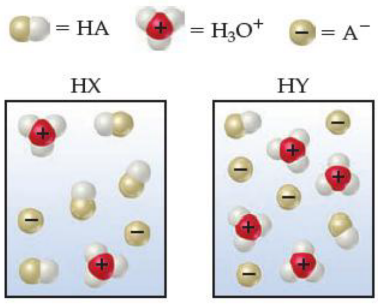

(a) HY.

(b). X⁻.

(c). left. $K_c < 1$.